

Juergen Gall

Background Subtraction and Tracking
Computer Vision
MA-INF 2201 / MA-MOROB-M04
WS25/26

Organisation

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- No lecture on 2.12. and 5.12.
- Next lecture on 9.12.
- 1st Exam: 24.02.2026: 12:30-14:30
Lecture hall 2, Friedrich-Hirzebruch-Allee 5, Hörsaalzentrum
Registration (HISinOne): 01.12.2025 - 21.12.2025
- 2nd Exam: 20.03.2026: 9:30-11:30
Lecture hall 2, Friedrich-Hirzebruch-Allee 5, Hörsaalzentrum
Registration (HISinOne): 01.12.2025 - 13.03.2026

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Grab Cut

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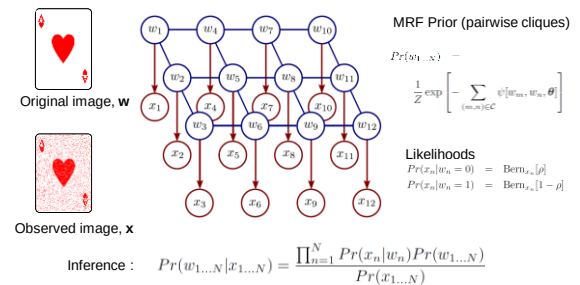
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Source: K. Grauman

Denosing with MRFs

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Grab Cut

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- Loosely specify foreground region
- Iterated graph cut



User initialization

 $Pr(x_n | w = j)$

Learning colour distributions

Graph cuts to infer the segmentation

C. Rother et al. GrabCut - Interactive Foreground Extraction using Iterated Graph Cuts. SIGGRAPH 2004

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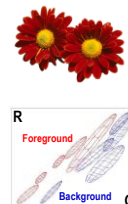
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Grab Cut

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- Loosely specify foreground region
- Iterated graph cut



Gaussian Mixture Model (typically 5-8 components)

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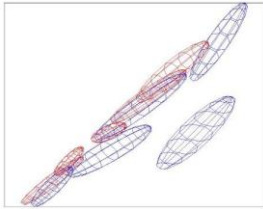
Source: K. Grauman

How do learn Gaussian mixture model?



Expectation-maximization:

$$Pr(\mathbf{x}_n | w = j) = \sum_{k=1}^K \lambda_{jk} \text{Norm}_{\mathbf{x}_n}[\boldsymbol{\mu}_{jk}, \boldsymbol{\Sigma}_{jk}]$$

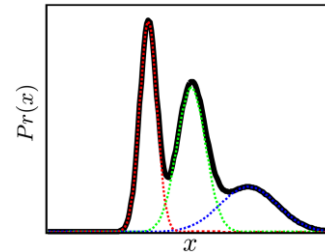


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Mixture of Gaussians (MoG)



$$Pr(\mathbf{x}|\boldsymbol{\theta}) = \sum_{k=1}^K \lambda_k \text{Norm}_{\mathbf{x}}[\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k]$$

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Source: S. Prince

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MoG as a marginalization



Define a variable $h \in \{1 \dots K\}$ and then write

$$Pr(\mathbf{x}|h, \boldsymbol{\theta}) = \text{Norm}_{\mathbf{x}}[\boldsymbol{\mu}_h, \boldsymbol{\Sigma}_h]$$

$$Pr(h|\boldsymbol{\theta}) = \lambda_k$$

Then we can recover the density by marginalizing $Pr(\mathbf{x}, h)$

$$\begin{aligned} Pr(\mathbf{x}|\boldsymbol{\theta}) &= \sum_{k=1}^K Pr(\mathbf{x}, h = k|\boldsymbol{\theta}) \\ &= \sum_{k=1}^K Pr(\mathbf{x}|h = k, \boldsymbol{\theta}) Pr(h = k|\boldsymbol{\theta}) \\ &= \sum_{k=1}^K \lambda_k \text{Norm}_{\mathbf{x}}[\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k]. \end{aligned}$$

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Source: S. Prince

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MoG as a marginalization



Define a variable $h \in \{1 \dots K\}$ and then write

$$Pr(\mathbf{x}|h, \boldsymbol{\theta}) = \text{Norm}_{\mathbf{x}}[\boldsymbol{\mu}_h, \boldsymbol{\Sigma}_h]$$

$$Pr(h|\boldsymbol{\theta}) = \lambda_k$$

Note:

- This gives us a method to generate data from MoG
First sample $Pr(h)$, then sample $Pr(\mathbf{x}|h)$
- The hidden variable h has a clear interpretation –
it tells you which Gaussian created data point $\mathbf{x}$

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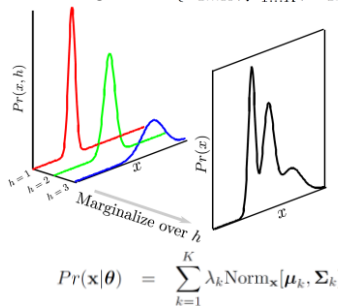
Source: S. Prince

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Expectation Maximization for MoG



How do we get $\boldsymbol{\theta} = \{\lambda_{1 \dots K}, \boldsymbol{\mu}_{1 \dots K}, \boldsymbol{\Sigma}_{1 \dots K}\}$?



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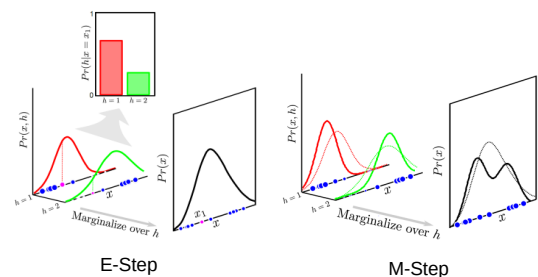
Source: S. Prince

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Expectation Maximization for MoG



- Iterate between Expectation (E) and Maximization (M)



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Source: S. Prince

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Expectation Maximization for MoG



GOAL: to learn parameters $\theta = \{\lambda_{1...K}, \mu_{1...K}, \Sigma_{1...K}\}$ from training data $\mathbf{x}_{1...I}$

E-Step – Maximize bound w.r.t. distributions $q(\mathbf{h})$

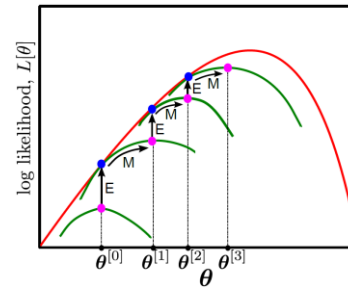
$$\hat{q}_i(\mathbf{h}_i) = Pr(\mathbf{h}_i | \mathbf{x}_i, \theta^{[t]}) = \frac{Pr(\mathbf{x}_i | \mathbf{h}_i, \theta^{[t]}) Pr(\mathbf{h}_i | \theta^{[t]})}{Pr(\mathbf{x}_i)}$$

M-Step – Maximize bound w.r.t. parameters θ

$$\hat{\theta}^{[t+1]} = \underset{\theta}{\operatorname{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K \hat{q}_i(\mathbf{h}_i = k) \log [Pr(\mathbf{x}_i, \mathbf{h}_i = k | \theta)] \right]$$

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Expectation Maximization



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Expectation Maximization for MoG



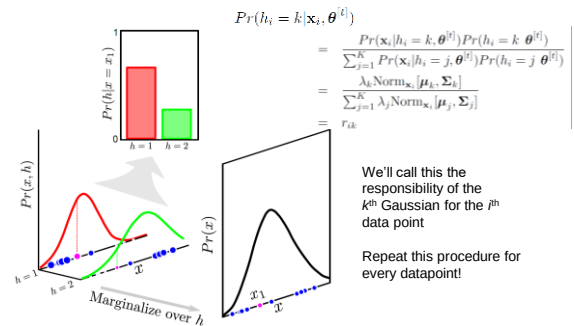
GOAL: to learn parameters $\theta = \{\lambda_{1...K}, \mu_{1...K}, \Sigma_{1...K}\}$ from training data $\mathbf{x}_{1...I}$

E-Step – Maximize bound w.r.t. distributions $q(\mathbf{h})$

$$\hat{q}_i(\mathbf{h}_i) = Pr(\mathbf{h}_i | \mathbf{x}_i, \theta^{[t]}) = \frac{Pr(\mathbf{x}_i | \mathbf{h}_i, \theta^{[t]}) Pr(\mathbf{h}_i | \theta^{[t]})}{Pr(\mathbf{x}_i)}$$

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E-Step



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Expectation Maximization for MoG



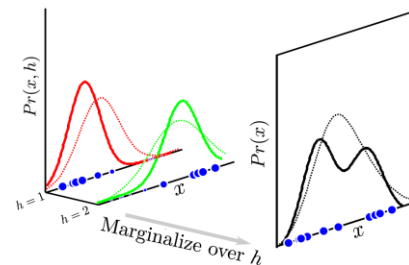
GOAL: to learn parameters $\theta = \{\lambda_{1...K}, \mu_{1...K}, \Sigma_{1...K}\}$ from training data $\mathbf{x}_{1...I}$

M-Step – Maximize bound w.r.t. parameters θ

$$\hat{\theta}^{[t+1]} = \underset{\theta}{\operatorname{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K \hat{q}_i(\mathbf{h}_i = k) \log [Pr(\mathbf{x}_i, \mathbf{h}_i = k | \theta)] \right]$$

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M-Step



Update means, covariances and weights according to responsibilities of datapoints

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M-Step

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$$\begin{aligned}\hat{\theta}^{[t+1]} &= \underset{\theta}{\operatorname{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K \hat{q}_i(h_i = k) \log [Pr(\mathbf{x}_i, h_i = k | \theta)] \right] \\ &= \underset{\theta}{\operatorname{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K r_{ik} \log [\lambda_k \operatorname{Norm}_{\mathbf{x}_i}[\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k]] \right].\end{aligned}$$

Take derivative, equate to zero and solve (Lagrange multipliers for λ)

$$\begin{aligned}\lambda_k^{[t+1]} &= \frac{\sum_{j=1}^I \sum_{i=1}^K r_{ijk}}{\sum_{j=1}^I \sum_{i=1}^K r_{ij}} \\ \boldsymbol{\mu}_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik} \mathbf{x}_i}{\sum_{i=1}^I r_{ik}} \\ \boldsymbol{\Sigma}_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k^{[t+1]})(\mathbf{x}_i - \boldsymbol{\mu}_k^{[t+1]})^T}{\sum_{i=1}^I r_{ik}}\end{aligned}$$

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Source: S. Prince

Derivatives

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Scalar x , vector $\mathbf{x}$, matrix $\mathbf{X}$:

$$\begin{aligned}y = f(x) \quad \frac{\partial f}{\partial x} & \\ y = f(\mathbf{x}) \quad \frac{\partial f}{\partial \mathbf{x}} &= \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_N} \right) \\ y = f(\mathbf{X}) \quad \frac{\partial f}{\partial \mathbf{X}} &= \begin{pmatrix} \frac{\partial f}{\partial x_{11}} & \dots & \frac{\partial f}{\partial x_{1N}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_{M1}} & \dots & \frac{\partial f}{\partial x_{MN}} \end{pmatrix} \\ y = f(\mathbf{x}) \quad \frac{\partial f}{\partial \mathbf{x}} &= \begin{pmatrix} \frac{\partial f}{\partial x_1} & \dots & \frac{\partial f}{\partial x_M} \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_1} & \dots & \frac{\partial f}{\partial x_N} \end{pmatrix}\end{aligned}$$

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Derivatives

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$$\begin{aligned}\frac{\partial \mathbf{x}^T \mathbf{a}}{\partial \mathbf{x}} &= \mathbf{a} \\ \frac{\partial \mathbf{a}^T \mathbf{x}}{\partial \mathbf{x}} &= \mathbf{a} \\ \frac{\partial \mathbf{a}^T \mathbf{X} \mathbf{b}}{\partial \mathbf{X}} &= \mathbf{a} \mathbf{b}^T \\ \frac{\partial \mathbf{a}^T \mathbf{X}^T \mathbf{b}}{\partial \mathbf{X}} &= \mathbf{b} \mathbf{a}^T \\ \frac{\partial \mathbf{b}^T \mathbf{X}^T \mathbf{c}}{\partial \mathbf{X}} &= \mathbf{X}(\mathbf{b} \mathbf{c}^T + \mathbf{c} \mathbf{b}^T) \\ \frac{\partial (\mathbf{B} \mathbf{x} + \mathbf{b})^T \mathbf{C} (\mathbf{D} \mathbf{x} + \mathbf{d})}{\partial \mathbf{x}} &= \mathbf{B}^T \mathbf{C} (\mathbf{D} \mathbf{x} + \mathbf{d}) + \mathbf{D}^T \mathbf{C}^T (\mathbf{B} \mathbf{x} + \mathbf{b}) \\ \frac{\partial \mathbf{x}^T \mathbf{B} \mathbf{x}}{\partial \mathbf{x}} &= (\mathbf{B} + \mathbf{B}^T) \mathbf{x} \\ \frac{\partial \mathbf{b}^T \mathbf{X}^T \mathbf{D} \mathbf{X} \mathbf{c}}{\partial \mathbf{X}} &= \mathbf{D}^T \mathbf{X} \mathbf{b} \mathbf{c}^T + \mathbf{D} \mathbf{X} \mathbf{c} \mathbf{b}^T \\ \frac{\partial (\mathbf{X} \mathbf{b} + \mathbf{c})^T \mathbf{D} (\mathbf{X} \mathbf{b} + \mathbf{c})}{\partial \mathbf{X}} &= (\mathbf{D} + \mathbf{D}^T)(\mathbf{X} \mathbf{b} + \mathbf{c}) \mathbf{b}^T.\end{aligned}$$

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M-Step

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$$\begin{aligned}\hat{\theta}^{[t+1]} &= \underset{\theta}{\operatorname{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K \hat{q}_i(h_i = k) \log [Pr(\mathbf{x}_i, h_i = k | \theta)] \right] \\ &= \underset{\theta}{\operatorname{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K r_{ik} \log [\lambda_k \operatorname{Norm}_{\mathbf{x}_i}[\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k]] \right].\end{aligned}$$

Take derivative, equate to zero and solve (Lagrange multipliers for λ)

$$\begin{aligned}\lambda_k^{[t+1]} &= \frac{\sum_{j=1}^I \sum_{i=1}^K r_{ijk}}{\sum_{j=1}^I \sum_{i=1}^K r_{ij}} \\ \boldsymbol{\mu}_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik} \mathbf{x}_i}{\sum_{i=1}^I r_{ik}} \\ \boldsymbol{\Sigma}_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k^{[t+1]})(\mathbf{x}_i - \boldsymbol{\mu}_k^{[t+1]})^T}{\sum_{i=1}^I r_{ik}}\end{aligned}$$

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Source: S. Prince

Lagrange multiplier

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- Optimize $\lambda_1, \dots, \lambda_K$

$$\operatorname{argmax}_{\lambda_1, \dots, \lambda_K} \left[\sum_{i=1}^I \sum_{k=1}^K r_{ik} \log [\lambda_k \operatorname{Norm}_{\mathbf{x}_i}[\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k]] \right]$$

- Subject to constraint:

$$\sum_{k=1}^K \lambda_k = 1$$

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Lagrange multiplier

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- Maximize $\lambda_1, \dots, \lambda_K$

$$f(\lambda_1, \dots, \lambda_K) = \sum_{i=1}^I \sum_{k=1}^K r_{ik} \log [\lambda_k \operatorname{Norm}_{\mathbf{x}_i}[\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k]]$$

- Subject to constraint:

$$\sum_{k=1}^K \lambda_k - 1 = 0$$

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Lagrange multiplier



- Rewrite without constraints using Lagrange multiplier λ :

$$f(\lambda_1, \dots, \lambda_K, \lambda) = \sum_i \sum_k r_{ik} \log(\lambda_k \mathcal{N}(x_i; \mu_k, \Sigma_k)) + \lambda \left(\sum_k \lambda_k - 1 \right)$$

- Compute gradients gives K+1 equations:

$$\frac{\partial}{\partial \lambda_k} \sum_i \sum_k r_{ik} \log(\lambda_k \mathcal{N}(x_i; \mu_k, \Sigma_k)) + \lambda \left(\sum_k \lambda_k - 1 \right) = 0 \quad \forall k$$

$$\frac{\partial}{\partial \lambda} \sum_i \sum_k r_{ik} \log(\lambda_k \mathcal{N}(x_i; \mu_k, \Sigma_k)) + \lambda \left(\sum_k \lambda_k - 1 \right) = 0$$

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Lagrange multiplier



- Compute gradients

$$\frac{\partial}{\partial \lambda_k} \sum_i \sum_k r_{ik} \log(\lambda_k \mathcal{N}(x_i; \mu_k, \Sigma_k)) + \lambda \left(\sum_k \lambda_k - 1 \right) = 0 \quad \forall k$$

$$\frac{\partial}{\partial \lambda} \sum_i \sum_k r_{ik} \log(\lambda_k \mathcal{N}(x_i; \mu_k, \Sigma_k)) + \lambda \left(\sum_k \lambda_k - 1 \right) = 0$$

- Gives K+1 equations:

$$\sum_i \frac{r_{ik}}{\lambda_k} + \lambda = 0 \quad \forall k$$

$$\sum_{k=1}^K \lambda_k - 1 = 0$$

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Lagrange multiplier



- Gives K+1 equations

$$\sum_i \frac{r_{ik}}{\lambda_k} + \lambda = 0 \quad \forall k \quad \sum_{k=1}^K \lambda_k - 1 = 0$$

- We therefore get by summing over all K equations:

$$\sum_k \sum_i r_{ik} = -\lambda \sum_k \lambda_k = -\lambda$$

$$\lambda_k = \frac{\sum_i r_{ik}}{\sum_k \sum_i r_{ik}}$$

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Proof



$$\frac{\partial}{\partial \mu_k} \left\{ \sum_i \sum_k r_{ik} \log \left(\frac{\lambda_k}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right) \right) \right\} = 0$$

$$\frac{\partial}{\partial \mu_k} \left\{ \sum_i r_{ik} \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right) \right\} = 0$$

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Derivatives



$$\begin{aligned} \frac{\partial \mathbf{x}^T \mathbf{a}}{\partial \mathbf{x}} &= \mathbf{a} & \frac{\partial \mathbf{b}^T \mathbf{x}^T \mathbf{X} \mathbf{c}}{\partial \mathbf{x}} &= \mathbf{X}(\mathbf{b} \mathbf{c}^T + \mathbf{c} \mathbf{b}^T) \\ \frac{\partial \mathbf{a}^T \mathbf{x}}{\partial \mathbf{x}} &= \mathbf{a} & \frac{\partial (\mathbf{B} \mathbf{x} + \mathbf{b})^T \mathbf{C} (\mathbf{D} \mathbf{x} + \mathbf{d})}{\partial \mathbf{x}} &= \mathbf{B}^T \mathbf{C} (\mathbf{D} \mathbf{x} + \mathbf{d}) + \mathbf{D}^T \mathbf{C}^T (\mathbf{B} \mathbf{x} + \mathbf{b}) \\ \frac{\partial \mathbf{a}^T \mathbf{X} \mathbf{b}}{\partial \mathbf{x}} &= \mathbf{a} \mathbf{b}^T & \frac{\partial \mathbf{x}^T \mathbf{B} \mathbf{x}}{\partial \mathbf{x}} &= (\mathbf{B} + \mathbf{B}^T) \mathbf{x} \\ \frac{\partial \mathbf{a}^T \mathbf{X}^T \mathbf{b}}{\partial \mathbf{x}} &= \mathbf{b} \mathbf{a}^T & \frac{\partial \mathbf{b}^T \mathbf{x}^T \mathbf{D} \mathbf{X} \mathbf{c}}{\partial \mathbf{x}} &= \mathbf{D}^T \mathbf{X} \mathbf{b} \mathbf{c}^T + \mathbf{D} \mathbf{X} \mathbf{c} \mathbf{b}^T \\ & & \frac{\partial (\mathbf{X} \mathbf{b} + \mathbf{c})^T \mathbf{D} (\mathbf{X} \mathbf{b} + \mathbf{c})}{\partial \mathbf{x}} &= (\mathbf{D} + \mathbf{D}^T) (\mathbf{X} \mathbf{b} + \mathbf{c}) \mathbf{b}^T \end{aligned}$$

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Proof



$$\frac{\partial}{\partial \mu_k} \left\{ \sum_i \sum_k r_{ik} \log \left(\frac{\lambda_k}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right) \right) \right\} = 0$$

$$\frac{\partial}{\partial \mu_k} \left\{ \sum_i r_{ik} \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right) \right\} = 0$$

$$\sum_i r_{ik} \Sigma_k^{-1} (x_i - \mu_k) = 0$$

$$\frac{\partial (\mathbf{B} \mathbf{x} + \mathbf{b})^T \mathbf{C} (\mathbf{D} \mathbf{x} + \mathbf{d})}{\partial \mathbf{x}} = \mathbf{B}^T \mathbf{C} (\mathbf{D} \mathbf{x} + \mathbf{d}) + \mathbf{D}^T \mathbf{C}^T (\mathbf{B} \mathbf{x} + \mathbf{b})$$

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Proof

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$$\begin{aligned} \frac{\partial}{\partial \mu_k} \left\{ \sum_i \sum_k r_{ik} \log \left(\frac{\lambda_k}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right) \right) \right\} &= 0 \\ \frac{\partial}{\partial \mu_k} \left\{ \sum_i r_{ik} \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right) \right\} &= 0 \\ \sum_i r_{ik} \Sigma_k^{-1} (x_i - \mu_k) &= 0 \\ \Sigma_k \Sigma_k^{-1} \sum_i r_{ik} x_i &= \Sigma_k \Sigma_k^{-1} \mu_k \sum_i r_{ik} \\ \mu_k &= \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}} \end{aligned}$$

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Proof

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$$\frac{\partial}{\partial \Sigma_k^{-1}} \left\{ \sum_i \sum_k r_{ik} \log \left(\frac{\lambda_k}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right) \right) \right\} = 0$$

$$\frac{\partial}{\partial \Sigma_k^{-1}} \left\{ \sum_i r_{ik} \frac{1}{2} \log \left(\frac{1}{|\Sigma_k|} \right) + r_{ik} \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right) \right\} = 0$$

Using:

$$x^T A x = \text{Tr}(A x x^T)$$

$$|A^{-1}| = \frac{1}{|A|}$$

We get:

$$\frac{\partial}{\partial \Sigma_k^{-1}} \left\{ \sum_i r_{ik} \log (|\Sigma_k^{-1}|) - r_{ik} \text{Tr} (\Sigma_k^{-1} B_k) \right\} = 0 \quad B_k = (x_i - \mu_k)(x_i - \mu_k)^T$$

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Proof

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$$\frac{\partial}{\partial \Sigma_k^{-1}} \left\{ \sum_i r_{ik} \log (|\Sigma_k^{-1}|) - r_{ik} \text{Tr} (\Sigma_k^{-1} B_k) \right\} = 0 \quad B_k = (x_i - \mu_k)(x_i - \mu_k)^T$$

Using:

$$\frac{\partial}{\partial A} \text{Tr}(AB) = B^T$$

$$\frac{\partial}{\partial A} \log(|A|) = (A^{-1})^T$$

We get:

$$\begin{aligned} \sum_i r_{ik} (\Sigma_k^{-1} - B_k^T) &= 0 \\ \sum_i r_{ik} (\Sigma_k - B_k) &= 0 \\ \Sigma_k &= \frac{\sum_i r_{ik} (x_i - \mu_k)(x_i - \mu_k)^T}{\sum_i r_{ik}} \end{aligned}$$

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M-Step

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$$\begin{aligned} \hat{\theta}^{[t+1]} &= \underset{\theta}{\text{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K \hat{q}_i(h_i = k) \log [Pr(\mathbf{x}_i, h_i = k | \theta)] \right] \\ &= \underset{\theta}{\text{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K r_{ik} \log [\lambda_k \text{Norm}_{\mathbf{x}_i}[\mu_k, \Sigma_k]] \right]. \end{aligned}$$

Take derivative, equate to zero and solve (Lagrange multipliers for λ)

$$\begin{aligned} \lambda_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik}}{\sum_{j=1}^K \sum_{i=1}^I r_{ij}} \\ \mu_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik} \mathbf{x}_i}{\sum_{i=1}^I r_{ik}} \\ \Sigma_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik} (\mathbf{x}_i - \mu_k^{[t+1]})(\mathbf{x}_i - \mu_k^{[t+1]})^T}{\sum_{i=1}^I r_{ik}} \end{aligned}$$

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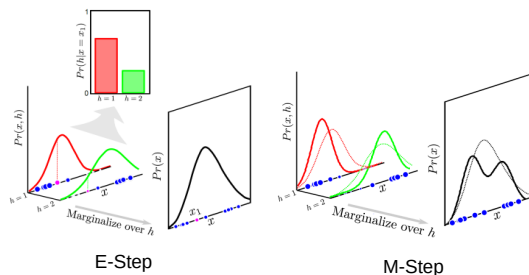
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Iterate until no further improvement

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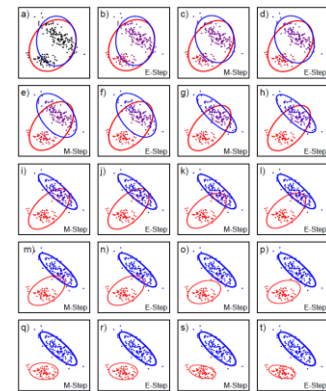


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Source: S. Prince

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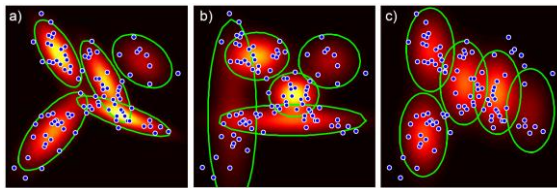
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Source: S. Prince

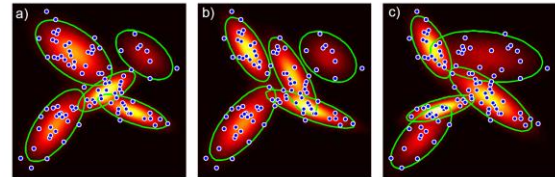
26

Different flavours...

Full
covarianceDiagonal
covarianceSame
covariance

Local Minima

Start from three random positions



L = 98.76

L = 96.97

L = 94.35

Expectation Maximization in General

Problem: Optimize cost functions of the form

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^I \log \left[\sum_h Pr(\mathbf{x}_i, \mathbf{h}_i) \right] \quad \leftarrow \text{Discrete case}$$

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^I \log \left[\int Pr(\mathbf{x}_i, \mathbf{h}_i) d\mathbf{h}_i \right] \quad \leftarrow \text{Continuous case}$$

Solution: Expectation Maximization (EM) algorithm
(Dempster, Laird and Rubin 1977)**Key idea:** Define lower bound on log-likelihood and increase at each iteration

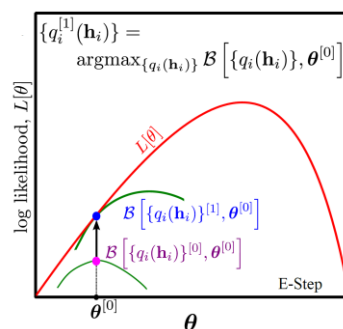
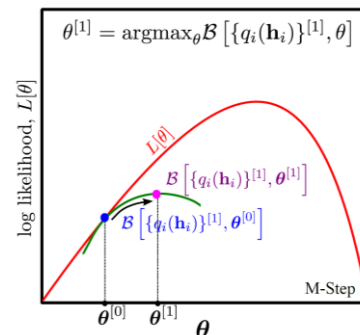
E-Step & M-Step

E-Step – Maximize bound w.r.t. distributions $\{q_i(\mathbf{h}_i)\}$

$$q_i^{[t]}[\mathbf{h}_i] = \arg \max_{q_i[\mathbf{h}_i]} \left[\mathcal{B}[\{q_i(\mathbf{h}_i)\}, \theta^{[t-1]}] \right]$$

M-Step – Maximize bound w.r.t. parameters θ

$$\theta^{[t]} = \arg \max_{\theta} \left[\mathcal{B}[\{q_i^{[t]}(\mathbf{h}_i)\}, \theta] \right]$$

**E-Step:** Update $\{q_i[\mathbf{h}_i]\}$ so that bound equals log likelihood for this**M-Step:** Update θ to maximum

Expectation Maximization

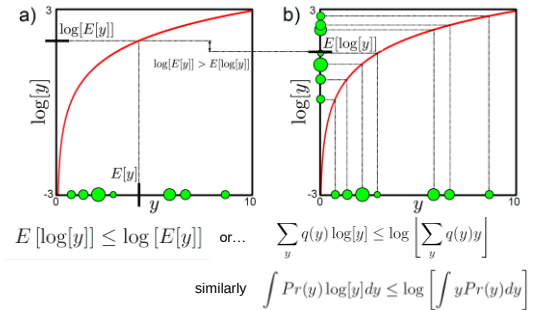


Defines a lower bound on log likelihood and increases bound iteratively

$$\begin{aligned}\mathcal{B}[\{q_i(\mathbf{h}_i)\}, \boldsymbol{\theta}] &= \sum_{i=1}^I \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{x}_i, \mathbf{h}_i | \boldsymbol{\theta})}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i \\ &\leq \sum_{i=1}^I \log \left[\int q_i(\mathbf{h}_i) \frac{Pr(\mathbf{x}_i, \mathbf{h}_i | \boldsymbol{\theta})}{q_i(\mathbf{h}_i)} d\mathbf{h}_i \right] \\ &= \sum_{i=1}^I \log \left[\int Pr(\mathbf{x}_i, \mathbf{h}_i | \boldsymbol{\theta}) d\mathbf{h}_i \right],\end{aligned}$$

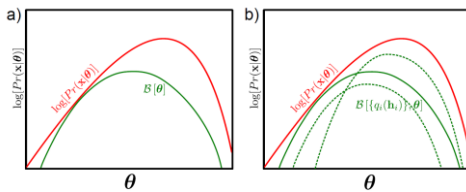
Source: S. Prince

Jensen's Inequality



Source: S. Prince

Lower bound



$$\mathcal{B}[\{q_i(\mathbf{h}_i)\}, \boldsymbol{\theta}] = \sum_{i=1}^I \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{x}_i, \mathbf{h}_i | \boldsymbol{\theta})}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i$$

Lower bound is a *function* of parameters $\boldsymbol{\theta}$ and a *set* of probability distributions $\{q_i(\mathbf{h}_i)\}$

Source: S. Prince

E-Step & M-Step



E-Step – Maximize bound w.r.t. distributions $\{q_i(\mathbf{h}_i)\}$

$$q_i^{[t]}[\mathbf{h}_i] = \operatorname{argmax}_{q_i[\mathbf{h}_i]} \left[\mathcal{B}[\{q_i(\mathbf{h}_i)\}, \boldsymbol{\theta}^{[t-1]}] \right]$$

M-Step – Maximize bound w.r.t. parameters $\boldsymbol{\theta}$

$$\boldsymbol{\theta}^{[t]} = \operatorname{argmax}_{\boldsymbol{\theta}} \left[\mathcal{B}[\{q_i^{[t]}(\mathbf{h}_i)\}, \boldsymbol{\theta}] \right]$$

Source: S. Prince

E-Step & M-Step



E-Step – Maximize bound w.r.t. distributions $q_i(\mathbf{h}_i)$

$$\hat{q}_i(\mathbf{h}_i) = Pr(\mathbf{h}_i | \mathbf{x}_i, \boldsymbol{\theta}^{[t]}) = \frac{Pr(\mathbf{x}_i | \mathbf{h}_i, \boldsymbol{\theta}^{[t]}) Pr(\mathbf{h}_i | \boldsymbol{\theta}^{[t]})}{Pr(\mathbf{x}_i)}$$

M-Step – Maximize bound w.r.t. parameters $\boldsymbol{\theta}$

$$\hat{\boldsymbol{\theta}}^{[t+1]} = \operatorname{argmax}_{\boldsymbol{\theta}} \left[\sum_{i=1}^I \int \hat{q}_i(\mathbf{h}_i) \log [Pr(\mathbf{x}_i, \mathbf{h}_i | \boldsymbol{\theta})] d\mathbf{h}_i \right]$$

Source: S. Prince

E-Step



E-Step – Maximize bound w.r.t. distributions $\{q_i(\mathbf{h}_i)\}$

$$q_i^{[t]}[\mathbf{h}_i] = \operatorname{argmax}_{q_i[\mathbf{h}_i]} \left[\mathcal{B}[\{q_i(\mathbf{h}_i)\}, \boldsymbol{\theta}^{[t-1]}] \right]$$

Analytical solution:

$$\hat{q}_i(\mathbf{h}_i) = Pr(\mathbf{h}_i | \mathbf{x}_i, \boldsymbol{\theta}^{[t]}) = \frac{Pr(\mathbf{x}_i | \mathbf{h}_i, \boldsymbol{\theta}^{[t]}) Pr(\mathbf{h}_i | \boldsymbol{\theta}^{[t]})}{Pr(\mathbf{x}_i)}$$

Source: S. Prince

E-Step – Optimize bound w.r.t $\{q_i(\mathbf{h}_i)\}$ 

$$\begin{aligned}
 \mathcal{B}(\{q_i(\mathbf{h}_i)\}, \theta) &= \sum_{i=1}^I \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{x}_i, \mathbf{h}_i | \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i \\
 &= \sum_{i=1}^I \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta) Pr(\mathbf{x}_i | \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i \\
 &= \sum_{i=1}^I \int q_i(\mathbf{h}_i) \log [Pr(\mathbf{x}_i | \theta)] d\mathbf{h}_i - \sum_{i=1}^I \int q_i(\mathbf{h}_i) \log \left[\frac{q_i(\mathbf{h}_i)}{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)} \right] d\mathbf{h}_i \\
 &= \sum_{i=1}^I \log [Pr(\mathbf{x}_i | \theta)] - \sum_{i=1}^I \int q_i(\mathbf{h}_i) \log \left[\frac{q_i(\mathbf{h}_i)}{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)} \right] d\mathbf{h}_i
 \end{aligned}$$

Constant w.r.t. $q(\mathbf{h})$

Only this term matters

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Source: S. Príncipe

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E-Step



$$\hat{q}_i(\mathbf{h}_i) = \operatorname{argmax}_{q_i(\mathbf{h}_i)} \left[- \int q_i(\mathbf{h}_i) \log \left[\frac{q_i(\mathbf{h}_i)}{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)} \right] d\mathbf{h}_i \right]$$

Kullback Leibler divergence – distance between probability distributions. We are maximizing the negative distance (i.e. Minimizing distance)

$$\hat{q}_i(\mathbf{h}_i) = Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)$$

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Source: S. Príncipe

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E-Step



$$\begin{aligned}
 \hat{q}_i(\mathbf{h}_i) &= \operatorname{argmax}_{q_i(\mathbf{h}_i)} \left[- \int q_i(\mathbf{h}_i) \log \left[\frac{q_i(\mathbf{h}_i)}{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)} \right] d\mathbf{h}_i \right] \\
 &= \operatorname{argmax}_{q_i(\mathbf{h}_i)} \left[\int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i \right] \\
 &= \operatorname{argmin}_{q_i(\mathbf{h}_i)} \left[- \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i \right]
 \end{aligned}$$

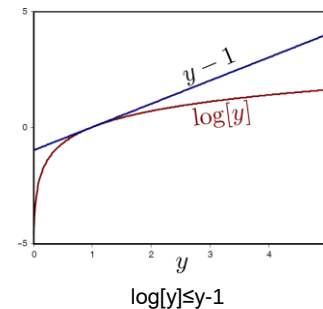
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Source: S. Príncipe

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Use this relation



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Source: S. Príncipe

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Kullback Leibler Divergence



$$\begin{aligned}
 \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i &\leq \int q_i(\mathbf{h}_i) \left(\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)}{q_i(\mathbf{h}_i)} - 1 \right) d\mathbf{h}_i \\
 &= \int Pr(\mathbf{h}_i | \mathbf{x}_i, \theta) - q_i(\mathbf{h}_i) d\mathbf{h}_i \\
 &= 1 - 1 = 0,
 \end{aligned}$$

So the cost function must be positive

$$\hat{q}_i(\mathbf{h}_i) = \operatorname{argmin}_{q_i(\mathbf{h}_i)} \left[- \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i \right]$$

In other words, the best we can do is choose $q_i(\mathbf{h}_i)$ so that this is zero

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Source: S. Príncipe

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E-Step



So the cost function must be positive

$$\hat{q}_i(\mathbf{h}_i) = \operatorname{argmin}_{q_i(\mathbf{h}_i)} \left[- \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i \right]$$

The best we can do is choose $q_i(\mathbf{h}_i)$ so that this is zero.

How can we do this? Easy – choose posterior $Pr(\mathbf{h} | \mathbf{x})$

$$\begin{aligned}
 \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_i &= \int Pr(\mathbf{h}_i | \mathbf{x}_i, \theta) \log \left[\frac{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)}{Pr(\mathbf{h}_i | \mathbf{x}_i, \theta)} \right] d\mathbf{h}_i \\
 &= \int Pr(\mathbf{h}_i | \mathbf{x}_i, \theta) \log [1] d\mathbf{h}_i = 0.
 \end{aligned}$$

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Source: S. Príncipe

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E-Step

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E-Step – Maximize bound w.r.t. distributions $\{q_i(\mathbf{h}_i)\}$

$$q_i^{[t]}[\mathbf{h}_i] = \operatorname{argmax}_{q_i[\mathbf{h}_i]} \left[\mathcal{B}[\{q_i(\mathbf{h}_i)\}, \theta^{[t-1]}] \right]$$

Analytical solution:

$$\hat{q}_i(\mathbf{h}_i) = Pr(\mathbf{h}_i | \mathbf{x}_i, \theta^{[t]}) = \frac{Pr(\mathbf{x}_i | \mathbf{h}_i, \theta^{[t]}) Pr(\mathbf{h}_i | \theta^{[t]})}{Pr(\mathbf{x}_i)}$$

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Source: S. Prince

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M-Step

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M-Step – Maximize bound w.r.t. parameters θ

$$\theta^{[t]} = \operatorname{argmax}_{\theta} \left[\mathcal{B}[\{q_i^{[t]}(\mathbf{h}_i)\}, \theta] \right]$$

Simplifies to:

$$\hat{\theta}^{[t+1]} = \operatorname{argmax}_{\theta} \left[\sum_{i=1}^I \int \hat{q}_i(\mathbf{h}_i) \log [Pr(\mathbf{x}_i, \mathbf{h}_i | \theta)] d\mathbf{h}_i \right]$$

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Source: S. Prince

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M-Step – Optimize bound w.r.t. θ

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$$\begin{aligned} \theta^{[t]} &= \operatorname{argmax}_{\theta} \left[\mathcal{B}[\{q_i^{[t]}(\mathbf{h}_i)\}, \theta] \right] \\ &= \operatorname{argmax}_{\theta} \left[\sum_{i=1}^I \int q_i^{[t]}(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{x}_i, \mathbf{h}_i | \theta)}{q_i^{[t]}(\mathbf{h}_i)} \right] d\mathbf{h}_i \right] \\ &= \operatorname{argmax}_{\theta} \left[\sum_{i=1}^I \int q_i^{[t]}(\mathbf{h}_i) \log [Pr(\mathbf{x}_i, \mathbf{h}_i | \theta)] - q_i^{[t]}(\mathbf{h}_i) \log [q_i^{[t]}(\mathbf{h}_i)] d\mathbf{h}_i \right] \\ &= \operatorname{argmax}_{\theta} \left[\sum_{i=1}^I \int q_i^{[t]}(\mathbf{h}_i) \log [Pr(\mathbf{x}_i, \mathbf{h}_i | \theta)] d\mathbf{h}_i \right] \end{aligned}$$

In the M-Step we optimize expected joint log likelihood with respect to parameters θ (Expectation w.r.t. distribution from E-Step)

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Source: S. Prince

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E-Step & M-Step

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E-Step – Maximize bound w.r.t. distributions $q_i(\mathbf{h}_i)$

$$\hat{q}_i(\mathbf{h}_i) = Pr(\mathbf{h}_i | \mathbf{x}_i, \theta^{[t]}) = \frac{Pr(\mathbf{x}_i | \mathbf{h}_i, \theta^{[t]}) Pr(\mathbf{h}_i)}{Pr(\mathbf{x}_i)}$$

M-Step – Maximize bound w.r.t. parameters θ

$$\hat{\theta}^{[t+1]} = \operatorname{argmax}_{\theta} \left[\sum_{i=1}^I \int \hat{q}_i(\mathbf{h}_i) \log [Pr(\mathbf{x}_i, \mathbf{h}_i | \theta)] d\mathbf{h}_i \right]$$

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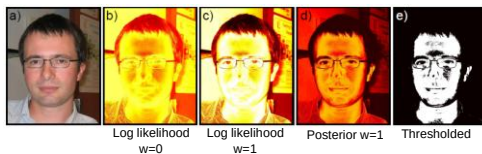
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Source: S. Prince

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Skin detection

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GMM for skin ($w=1$) and non-skin ($w=0$):

$$Pr(x|w=k) = \sum_i \lambda_{k,i} \operatorname{Norm}_x[\mu_{k,i}, \Sigma_{k,i}] \quad Pr(w) = \operatorname{Bern}_w[\lambda]$$

$$Pr(w=1|x) = \frac{Pr(x|w=1)Pr(w=1)}{\sum_{k=0}^1 Pr(x|w=k)Pr(w=k)}$$

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Human pose estimation

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[M. Dantone et al. Body Parts Dependent Joint Regressors for Human Pose Estimation in Still Images. PAMI 2014]

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Background subtraction



Model background distribution of each pixel by GMM:

$$Pr(x_n | w = 0) = \sum_i \lambda_{n,i} \text{Norm}_{x_n}[\mu_{n,i}, \Sigma_{n,i}]$$

$$Pr(x_n | w = 1) = \kappa_i$$



Test image

Pixel-wise classification

MRF

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Background subtraction



Shadows → Separate chromaticity from intensity (two thresholds)

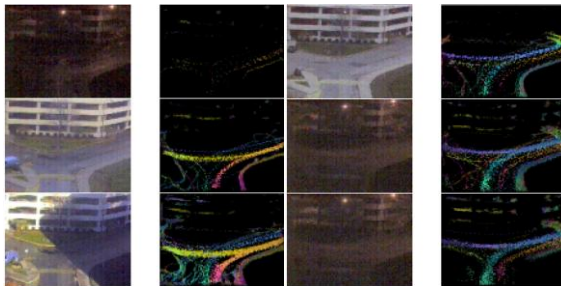


Examples: HSV, LAB, rgs, ... $r = \frac{R}{R+G+B}, g = \frac{G}{R+G+B}, b = \frac{B}{R+G+B}$

A. Elgammal. **Figure-ground segmentation - pixel-based**. Springer 2011

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Cars and pedestrians



1 day

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Background adaptation



MoG:

$$P(X_t) = \sum_{i=1}^K \omega_{i,t} * \eta(X_t, \mu_{i,t}, \Sigma_{i,t})$$

Update over time:

$$\omega_{k,t} = (1 - \alpha)\omega_{k,t-1} + \alpha(M_{k,t}) \quad M_{k,t} = \begin{cases} 1 & \|X_t - \mu_k\| < 2.5\sigma_k \\ 0 & \text{otherwise} \end{cases}$$

$$\mu_t = (1 - \rho)\mu_{t-1} + \rho X_t$$

$$\sigma_t^2 = (1 - \rho)\sigma_{t-1}^2 + \rho(X_t - \mu_t)^T(X_t - \mu_t)$$

Distributions that are more likely and have less variance are usually part of the background.

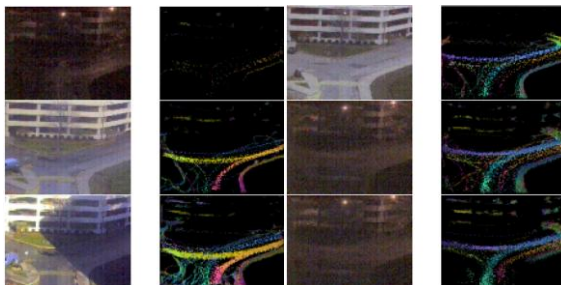
Sort Gaussians decreasingly by ω/σ to obtain background model:

$$B = \text{argmin}_b \left(\sum_{k=1}^b \omega_k > T \right)$$

C. Stauffer and W. Grimson. **Adaptive background mixture models for real-time tracking**. CVPR 1999

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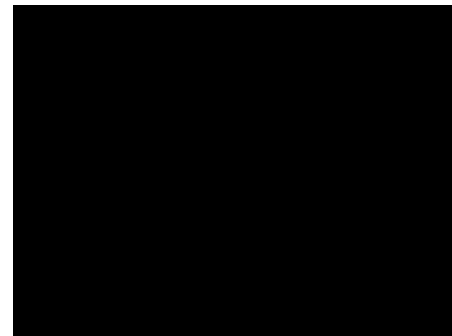
Cars and pedestrians



1 day

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Human pose tracking: 1997



S. Wren et al. **Pfinder: Real-Time Tracking of the Human Body**. TPAMI 1997

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Mean Shift Tracking

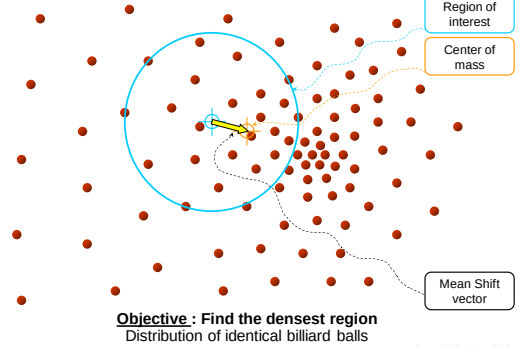
Goal: Mark object in first frame and locate it in all frames



D. Comaniciu. Real-Time Tracking of Non-Rigid Objects using Mean Shift. CVPR 2000

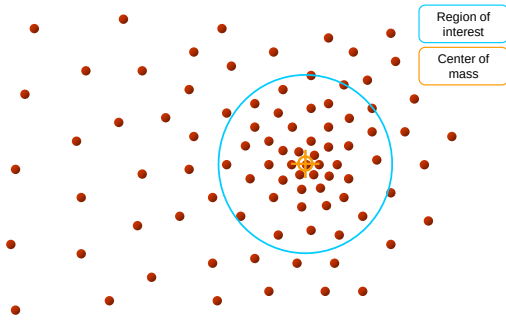
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Recall: Mean Shift



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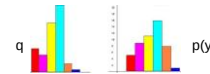
Recall: Mean Shift



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Template

Distance between template q (histogram of color) and target $p(y)$ (histogram of color), given by Bhattacharyya coefficient:



$$d(y) = \sqrt{1 - \rho[\hat{p}(y), \hat{q}]}$$

$$\hat{\rho}(y) \equiv \rho[\hat{p}(y), \hat{q}] = \sum_{u=1}^m \sqrt{\hat{p}_u(y) \hat{q}_u}$$

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Mean shift

Mean shift for x_i data points:

$$y_{j+1} = \frac{\sum_{i=1}^n x_i g\left(\left\|\frac{y_j - x_i}{h}\right\|^2\right)}{\sum_{i=1}^n g\left(\left\|\frac{y_j - x_i}{h}\right\|^2\right)}$$

$$g(x) = -k'(x)$$

Look at pixel around y : x_i with histogram bin $b(x_i)$ to get histogram $\sum_{u=1}^m \hat{p}_u = 1$



$$\hat{p}_u(y) = C_h \sum_{i=1}^{n_h} k\left(\left\|\frac{y - x_i}{h}\right\|^2\right) \delta[b(x_i) - u]$$

$$C_h = \frac{1}{\sum_{i=1}^{n_h} k\left(\left\|\frac{y - x_i}{h}\right\|^2\right)}$$

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Mean shift tracking

Maximize Bhattacharyya coefficient:

$$\hat{\rho}(y) \equiv \rho[\hat{p}(y), \hat{q}] = \sum_{u=1}^m \sqrt{\hat{p}_u(y) \hat{q}_u}$$

$$f(a) + \frac{f'(a)}{1!}(x - a)$$

Taylor expansion around $p(y_0)$:

$$\rho[\hat{p}(y), \hat{q}] \approx \frac{1}{2} \sum_{u=1}^m \sqrt{\hat{p}_u(\tilde{y}_0) \hat{q}_u} + \frac{1}{2} \sum_{u=1}^m \hat{p}_u(y) \sqrt{\frac{\hat{q}_u}{\hat{p}_u(\tilde{y}_0)}} \quad \hat{p}_u(\tilde{y}_0) > 0$$

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Mean shift tracking



Maximize Bhattacharyya coefficient:

$$\hat{\rho}(\mathbf{y}) \equiv \rho[\hat{\mathbf{p}}(\mathbf{y}), \hat{\mathbf{q}}] = \sum_{u=1}^m \sqrt{\hat{p}_u(\mathbf{y}) \hat{q}_u}$$

Taylor expansion around $\mathbf{p}(\mathbf{y}_0)$:

$$\rho[\hat{\mathbf{p}}(\mathbf{y}), \hat{\mathbf{q}}] \approx \frac{1}{2} \sum_{u=1}^m \sqrt{\hat{p}_u(\hat{\mathbf{y}}_0) \hat{q}_u} + \frac{1}{2} \sum_{u=1}^m \hat{p}_u(\mathbf{y}) \sqrt{\frac{\hat{q}_u}{\hat{p}_u(\hat{\mathbf{y}}_0)}} \quad \hat{p}_u(\hat{\mathbf{y}}_0) > 0$$

Using

$$\hat{p}_u(\mathbf{y}) = C_h \sum_{i=1}^{n_h} k\left(\left\|\frac{\mathbf{y} - \mathbf{x}_i}{h}\right\|^2\right) \delta[b(\mathbf{x}_i) - u]$$

We get:

$$\rho[\hat{\mathbf{p}}(\mathbf{y}), \hat{\mathbf{q}}] \approx \frac{1}{2} \sum_{u=1}^m \sqrt{\hat{p}_u(\hat{\mathbf{y}}_0) \hat{q}_u} + \frac{C_h}{2} \sum_{i=1}^{n_h} w_i k\left(\left\|\frac{\mathbf{y} - \mathbf{x}_i}{h}\right\|^2\right) \quad w_i = \sum_{u=1}^m \delta[b(\mathbf{x}_i) - u] \sqrt{\frac{\hat{q}_u}{\hat{p}_u(\hat{\mathbf{y}}_0)}}$$

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Mean shift tracking



Start with position from previous frame $\mathbf{y}_0$ and template $\mathbf{q}$

$$\rho[\hat{\mathbf{p}}(\hat{\mathbf{y}}_0), \hat{\mathbf{q}}] = \sum_{u=1}^m \sqrt{\hat{p}_u(\hat{\mathbf{y}}_0) \hat{q}_u}$$

Compute weights:

$$w_i = \sum_{u=1}^m \delta[b(\mathbf{x}_i) - u] \sqrt{\frac{\hat{q}_u}{\hat{p}_u(\hat{\mathbf{y}}_0)}}$$

Mean shift:

$$\hat{\mathbf{y}}_1 = \frac{\sum_{i=1}^{n_h} \mathbf{x}_i w_i g\left(\left\|\frac{\hat{\mathbf{y}}_0 - \mathbf{x}_i}{h}\right\|^2\right)}{\sum_{i=1}^{n_h} w_i g\left(\left\|\frac{\hat{\mathbf{y}}_0 - \mathbf{x}_i}{h}\right\|^2\right)}$$

Update:

$$\rho[\hat{\mathbf{p}}(\hat{\mathbf{y}}_1), \hat{\mathbf{q}}] = \sum_{u=1}^m \sqrt{\hat{p}_u(\hat{\mathbf{y}}_1) \hat{q}_u}$$

$$\text{While } \rho[\hat{\mathbf{p}}(\hat{\mathbf{y}}_1), \hat{\mathbf{q}}] < \rho[\hat{\mathbf{p}}(\hat{\mathbf{y}}_0), \hat{\mathbf{q}}]$$

$$\text{Do } \hat{\mathbf{y}}_1 \leftarrow \frac{1}{2} (\hat{\mathbf{y}}_0 + \hat{\mathbf{y}}_1).$$

$$\|\hat{\mathbf{y}}_1 - \hat{\mathbf{y}}_0\| < \epsilon$$

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Mean shift tracking



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Mean shift tracking

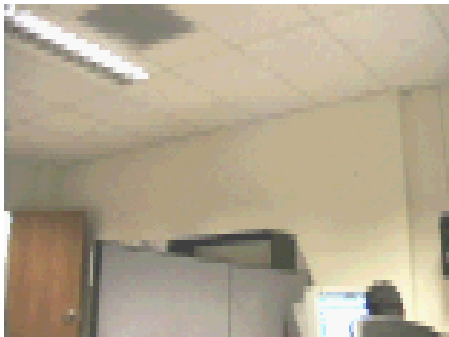


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Mean shift tracking



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Mean shift tracking



- Simple and fast
- Initialization matters
- Get stuck in local optima
- Temporal information is ignored
- Cannot handle occlusions

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